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Studierichting: Informatica
Oefeningenreeks 3D nr. 5

$$f(x) = \frac{3x^2+ax+6}{3x^2+bx+9}$$

$$f'(x) = \frac{(3x^2+ax+6)' \cdot (3x^2+bx+9) - (3x^2+ax+6) \cdot (3x^2+bx+9)'}{(3x^2+bx+9)^2}$$

$$= \frac{(6x+a) \cdot (3x^2+bx+9) - (3x^2+ax+6) \cdot (6x+b)}{(3x^2+bx+9)^2}$$

$$f'(x) = 0$$

$$\frac{(6x+a) \cdot (3x^2+bx+9) - (3x^2+ax+6) \cdot (6x+b)}{(3x^2+bx+9)^2} = 0$$

$$(6x+a) \cdot (3x^2+bx+9) - (3x^2+ax+6) \cdot (6x+b) = 0$$

$$f'(1) = 0$$

$$(6+a) \cdot (3+b+9) - (3+a+6) \cdot (6+b) = 0$$

$$\Leftrightarrow (6+a)(3+b+9) - (3+a+6)(6+b) = 0$$

$$\Leftrightarrow (6+a)(12+b) - (9+a)(6+b) = 0$$

$$\Leftrightarrow (72+6b+12a+ab) - (54+9b+6a+ab) = 0$$

$$\Leftrightarrow 72+6b+12a+ab-54-9b-6a-ab = 0$$

$$\Leftrightarrow 18+6a-3b = 0$$

$$\Leftrightarrow 6+2a-b = 0$$

$$\Leftrightarrow 2a-b = -6$$

$$f'(2) = 0$$

$$(6 \cdot 2+a) \cdot (3 \cdot 2^2+b \cdot 2+9) - (3 \cdot 2^2+a \cdot 2+6) \cdot (6 \cdot 2+b) = 0$$

$$\Leftrightarrow (12+a)(12+2b+9) - (12+2a+6)(12+b) = 0$$

$$\Leftrightarrow (12+a)(21+2b) - (18+2a)(12+b) = 0$$

$$\Leftrightarrow (252+24b+21a+2ab) - (216+18b+24a+2ab) = 0$$

$$\Leftrightarrow 252+24b+21a+2ab-216-18b-24a-2ab) = 0$$

$\Leftrightarrow 36 - 3a + 6b = 0$

$\Leftrightarrow 3a - 6b = 36$

$\Leftrightarrow a - 2b = 12$

$$\begin{cases} 2a - b = -6 \\ a - 2b = 12 \end{cases}$$

$$\begin{cases} b = 6 + 2a \\ a - 6 \cdot (6 + 2a) = 12 \end{cases}$$

$$\begin{cases} b = 6 + 2a \\ a - 12 - 4a = 12 \end{cases}$$

$$\begin{cases} b = 6 + 2a \\ a - 4a = 24 \end{cases}$$

$$\begin{cases} b = 6 + 2a \\ -3a = 24 \end{cases}$$

$$\begin{cases} b = 6 + 2 \cdot (-8) \\ a = -8 \end{cases}$$

$$\begin{cases} b = -10 \\ a = -8 \end{cases}$$

References